# Ciaran Regan

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## Education

**University of Tsukuba** Expected March 2028
*PhD in Computer Science* *Tsukuba, Japan*

- **Research Interests:** Biologically-inspired neural architectures, neural plasticity, artificial life systems.
- **Scholarship:** Japanese Government Monbukagakusho (MEXT) Scholarship

**University of Tsukuba** March 2025
*MSc in Computer Science (1st Class Honours)* *Tsukuba, Japan*

- **Scholarship:** Japanese Government Monbukagakusho (MEXT) Scholarship

**Trinity College Dublin** October 2021
*Bachelor of Arts in Theoretical Physics (1st Class Honours)* *Dublin, Ireland*

- **Scholarship:** All Ireland Scholarship

## Experience

**Sakana AI** October 2024 – Present
*Research Intern* *Tokyo, Japan*

- Founding member of Sakana AI's Frontier Intelligence Group (FIG).
- Researching biologically inspired recurrent architectures, including the Continuous Memory Machine and Continuous Thought Machine (NeurIPS 2025 Spotlight).
- Published a first-author survey paper on plastic and fast-weight recurrent neural networks in TMLR.
- Leading the Sakana AI research reading group.

**Freelance Research Assistant** April 2023 – September 2024
*Research supervised by Prof. Mizuki Oka* *Japan*

- Developed neural point process models for event time series prediction and collaborated with industry partners on integrating probabilistic programming methods into production systems.
- Built agentic modelling and probabilistic planning tools in Julia and Python using GPT-4 for social network analysis and online goal inference.

**Accenture** July 2021 – October 2022
*Data Scientist* *Dublin, Ireland*

- Built Scala/Spark network analytics for anti-money laundering at an international bank.
- Built ETL pipelines to construct knowledge graphs from banking and watchlist data.
- Trained European government staff on Scala and Apache Spark programming.

## Publications

**Differentiable Plastic Recurrent Neural Networks: A Systematic Evaluation**
**Ciaran Regan**, Kai Arulkumaran, Sebastian Risi, Llion Jones. *TMLR 2026.*

**Continuous Memory Machines**
**Ciaran Regan**, Kai Arulkumaran, Luke Darlow, Stefania Druga, Sebastian Risi, Llion Jones. *NeurIPS 2026 Workshop: PALM; under review at ICLR.*

**Microcosmos: Reimagining Artificial Life for the GPU Era**
Mark Tensen, **Ciaran Regan**, Bert Wang-Chak Chan, Mizuki Oka, Kenneth O. Stanley, Grisha Szep. *ALIFE 2026.*

**Discovering Novel LLM Experts via Task-Capability Coevolution**
Andrew Dai, Boris Meinardus, **Ciaran Regan**, Yingtao Tian, Yujin Tang. *ICLR 2026.*

**Continuous Thought Machines**
Luke Darlow, **Ciaran Regan**, Sebastian Risi, Jeffrey Seely, Llion Jones. *NeurIPS 2025 (Spotlight).*

**Problem-Solving in Language Model Networks**
**Ciaran Regan**, Alexandre Gournail, Mizuki Oka. *ALIFE 2024*.

**LLM-POET: Evolving Complex Environments using Large Language Models**
Fuma Aki, Riku Ikeda, Takumi Saito, **Ciaran Regan**, Mizuki Oka. *GECCO 2024*.

## Skills

**Languages**: English (native), Japanese (JLPT N4)
**Programming**: Python, JAX, PyTorch, Scala, Julia, Rust